

DHCP Server Installation and Scope Configuration Lab

Lab Overview

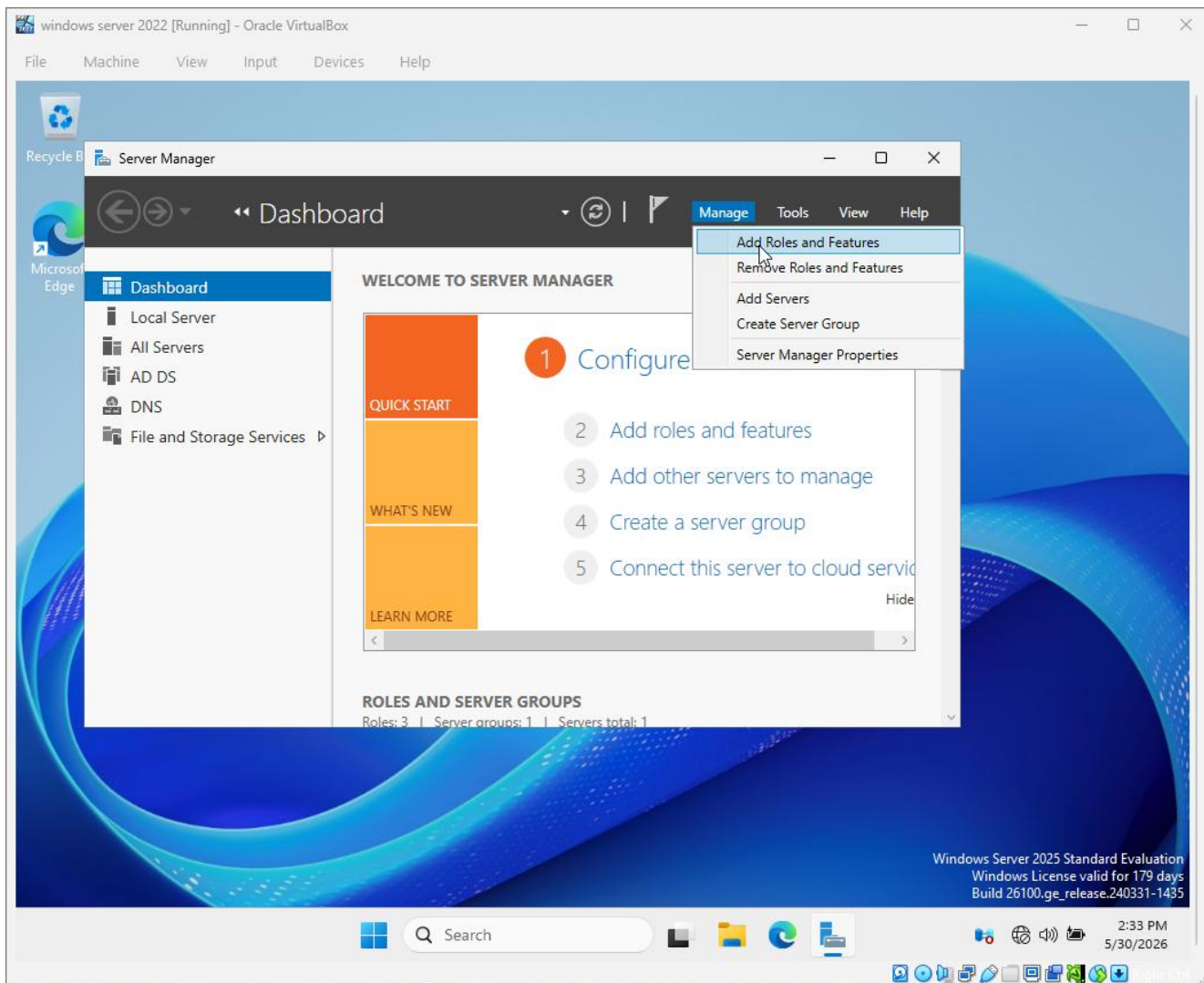
This hands-on Windows Server 2022 lab focused on installing and configuring the DHCP Server role, creating DHCP scopes, configuring gateway and DNS settings, and managing IP address allocation.

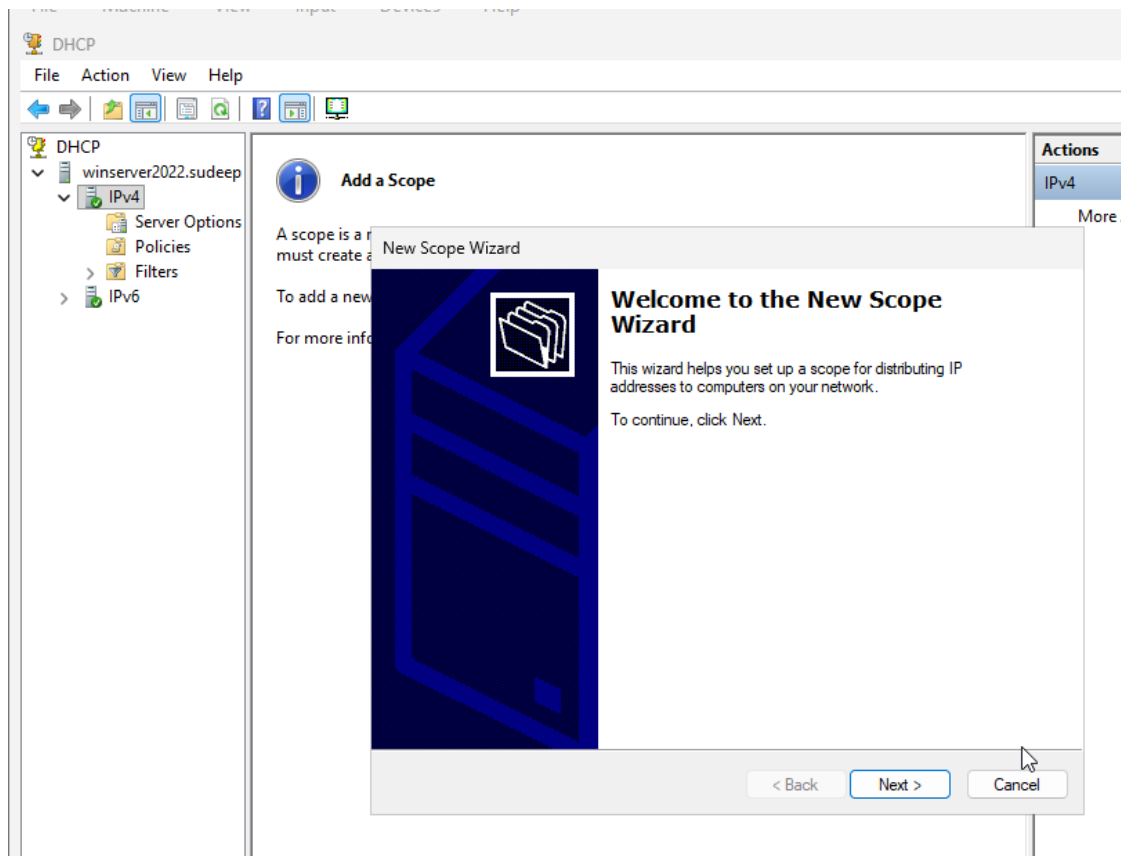
Topics Covered

- DHCP Server Installation
- DHCP Server Authorization
- DHCP Console Management
- DHCP Scope Creation
- IPv4 Scope Configuration
- IP Address Range Configuration
- Subnet Mask Configuration
- DHCP Lease Duration Configuration
- Default Gateway Configuration
- DNS Server Configuration
- Scope Options Configuration
- Active Directory Integration
- DHCP Administration
- Network Address Management

Conclusion

Successfully deployed and configured a DHCP Server in Windows Server 2022, including scope creation, gateway and DNS integration, and centralized IP address management.







Add a Scope

A scope is a...
must create a...

To add a new...

For more info...

New Scope Wizard

Scope Name

You have to provide an identifying scope name. You also have the option of providing a description.



Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

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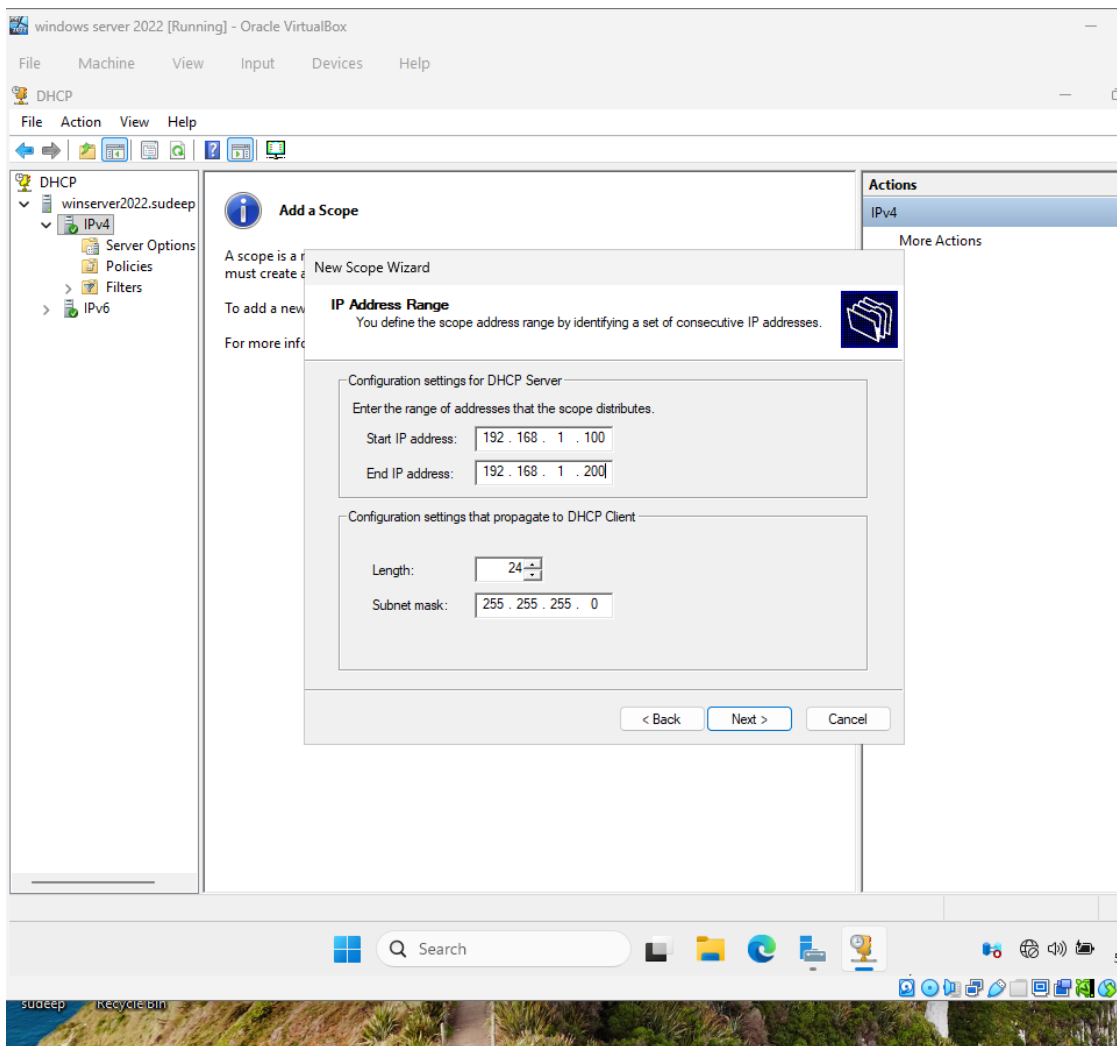
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Cancel

IPv4

IPv4

More Actions



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Add a Scope

A scope is a range of IP addresses that you must create a scope for.

To add a new scope, click the Add Scope button.

For more information, see the DHCP console help.

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.



Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address:

End IP address:

Add

Excluded address range:

Remove

Subnet delay in milli second:

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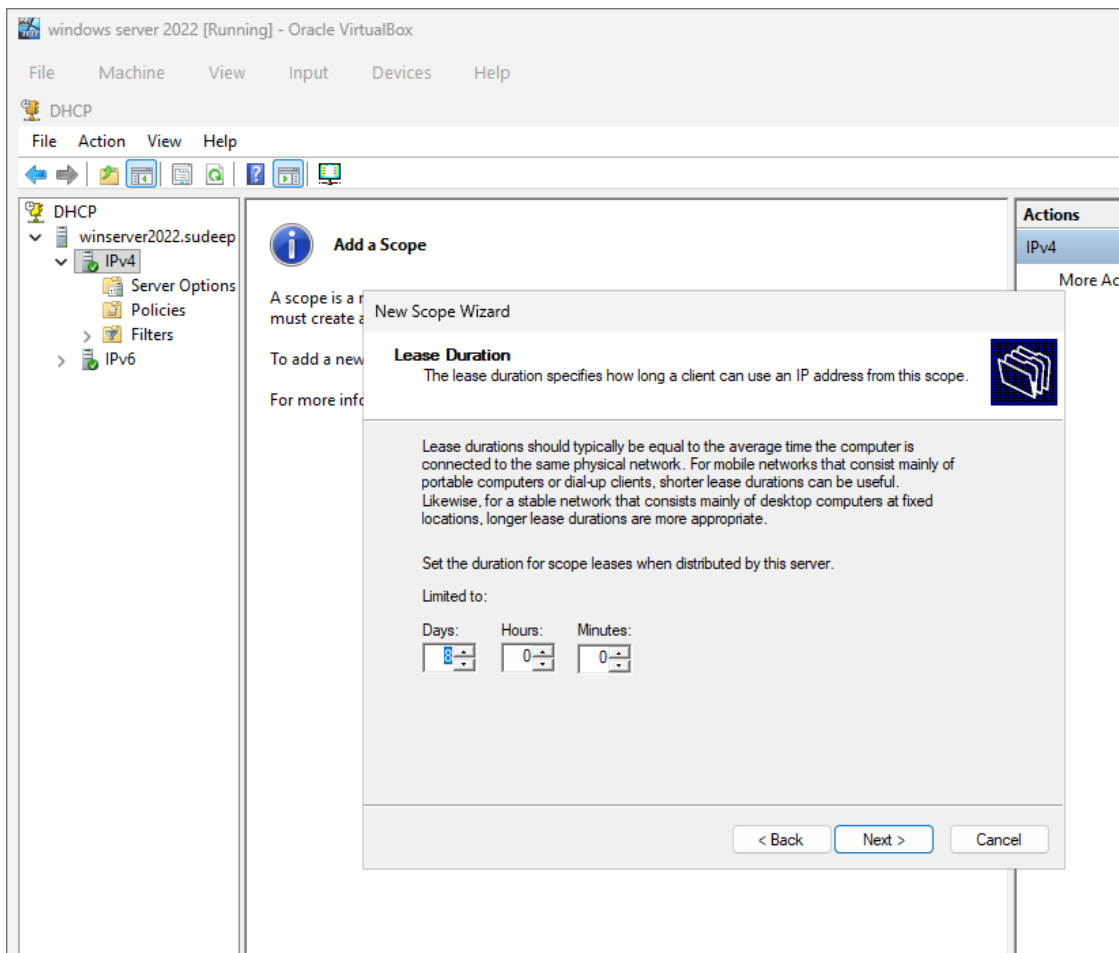
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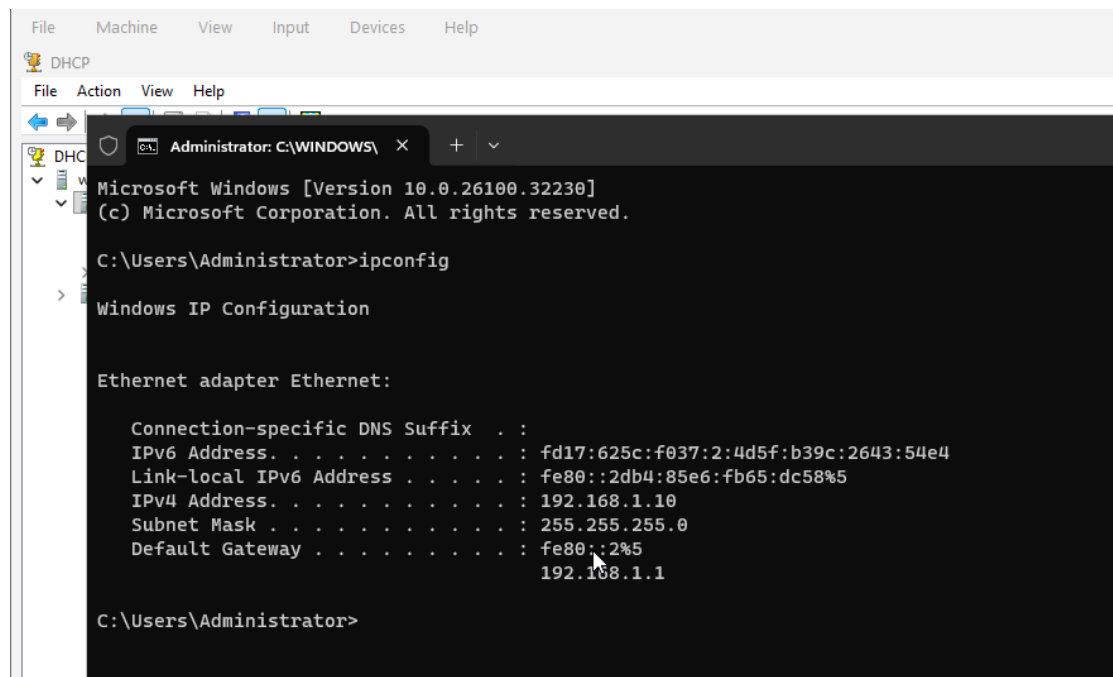
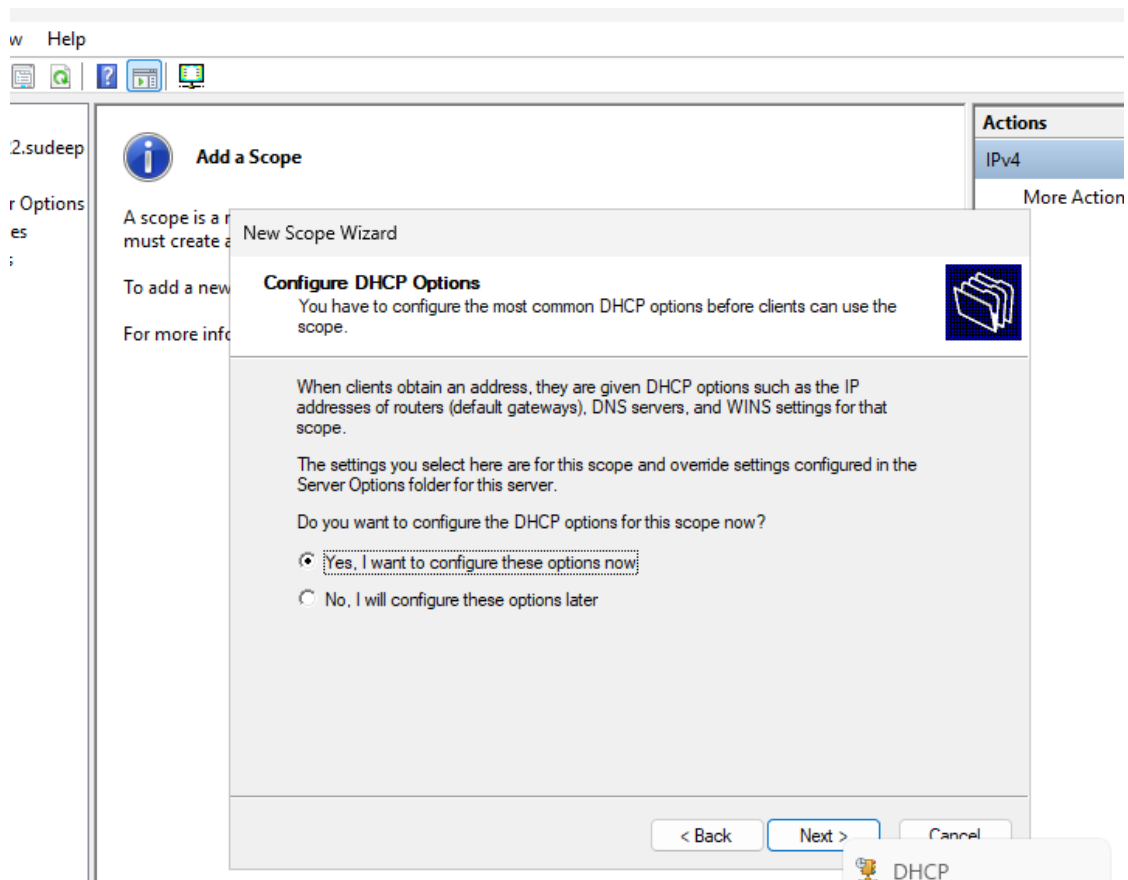
Cancel

Actions

IPv4

More Actions







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rs**Add a Scope**

A scope is a
must create a

To add a new

For more info

Actions

IPv4

More A

New Scope Wizard**Router (Default Gateway)**

You can specify the routers, or default gateways, to be distributed by this scope.



To add an IP address for a router used by clients, enter the address below.

IP address:

Add

192.168.1.1

Remove

Up

Down

< Back

Next >

Cancel



New Scope Wizard

WINS Servers

Computers running Windows can use WINS servers to convert NetBIOS computer names to IP addresses.



Entering server IP addresses here enables Windows clients to query WINS before they use broadcasts to register and resolve NetBIOS names.

Server name:

Resolve

IP address:

Add

Remove

Up

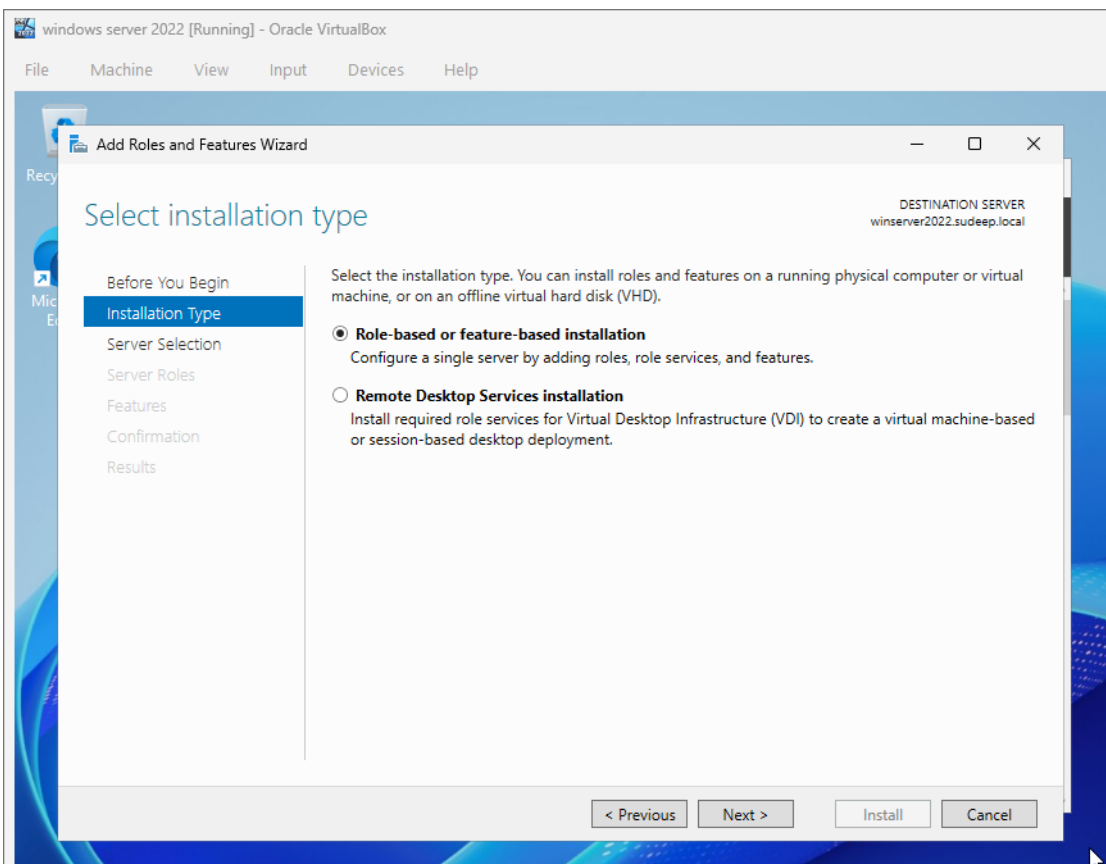
Down

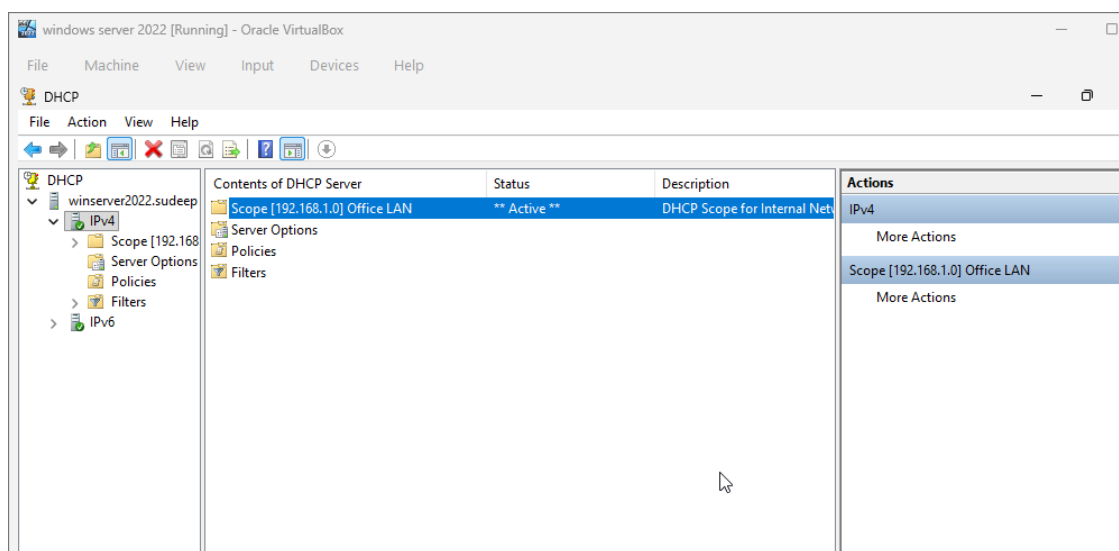
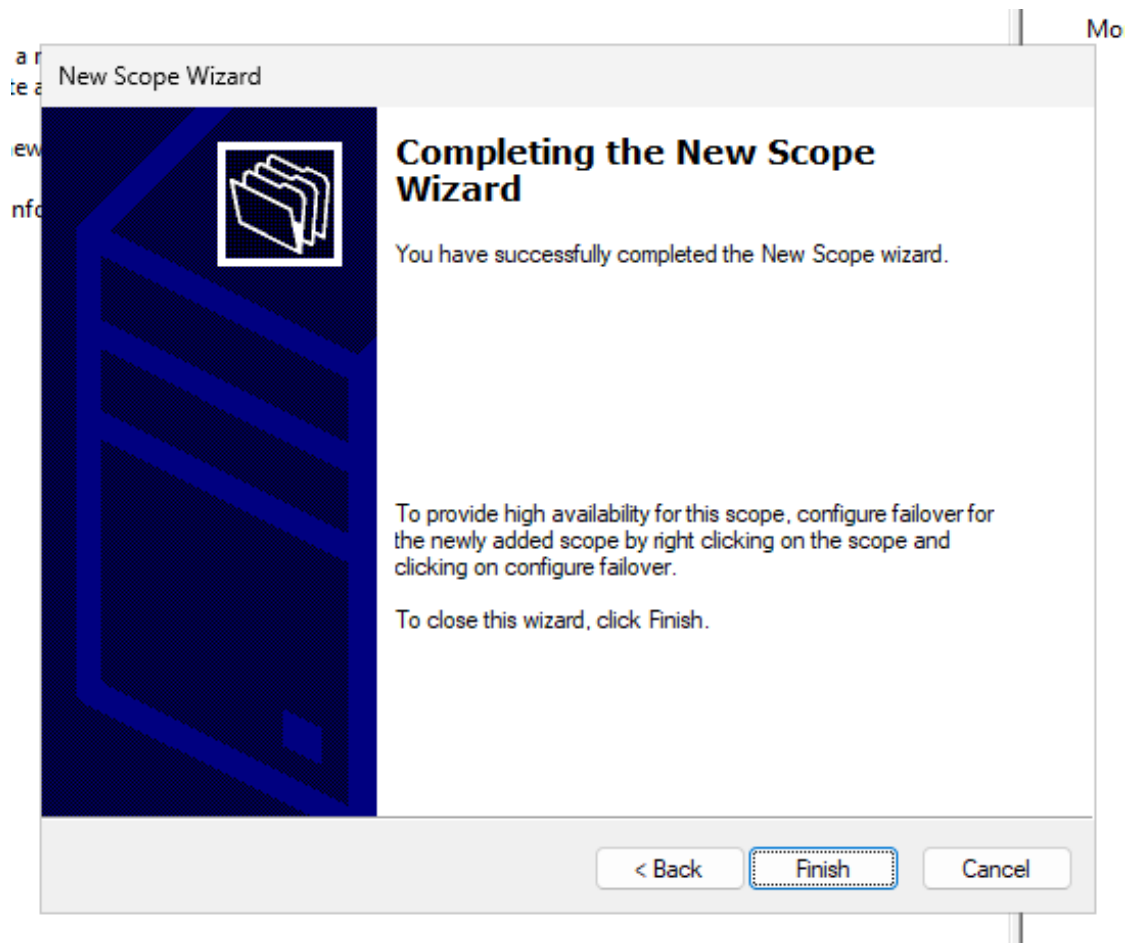
To change this behavior for Windows DHCP clients modify option 046, WINS/NBT Node Type, in Scope Options.

< Back

Next >

Cancel





windows server 2022 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

DHCP

File Action View Help

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DHCP

- winserver2022.sudeep
 - IPv4
 - Scope [192.168]
 - Address Po
 - Address Lei
 - > Reservation
 - Scope Opti
 - Policies
 - Server Options
 - Policies
 - > Filters
 - IPv6

Contents of Scope

- Address Pool
- Address Leases
- Reservations
- Scope Options
- Policies

Add Roles and Features Wizard

Recy

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Select server roles

DESTINATION SERVER
winserver2022.sudeep.local

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select one or more roles to install on the selected server.

Roles

☐ Active Directory Certificate Services

☒ Active Directory Domain Services (Installed)

☐ Active Directory Federation Services

☐ Active Directory Lightweight Directory Services

☐ Active Directory Rights Management Services

☐ Device Health Attestation

☐ DHCP Server

☒ DNS Server (Installed)

☐ Fax Server

☒ File and Storage Services (2 of 12 installed)

☐ Host Guardian Service

☐ Hyper-V

☐ Network Policy and Access Services

☐ Print and Document Services

☐ Remote Access

☐ Remote Desktop Services

☐ Volume Activation Services

☐ Web Server (IIS)

☐ Windows Deployment Services

☐ Windows Server Update Services

Description

Dynamic Host Configuration Protocol (DHCP) Server enables you to centrally configure, manage, and provide temporary IP addresses and related information for client computers.

< Previous

Next >

Install

Cancel

